

sharpen the focus on to individual subjects. Depth of field decreases with lower f-numbers.

Aperture width and shutter speed are inextricably linked. Here we take a look at the importance of shutter speed.

2.3 Shutter Speed

In a digital camera, the shutter serves to adjust the time during which the image sensor collects the light that it receives through the lens. Thus, along with determining the level of brightness of the image, the shutter manages to capture the image at a particular moment during which movement is negligible – almost freezing the action to catch a slice of space and time. A *shutter* usually stays shut (to keep the back of the camera – the image sensor – dark and unexposed) till you click, which is when it opens for a fraction of a second (anything from 4 to 1/4000th of a second) and then shuts again.

If you thought that the shutter is called a *shutter* because it shuts a lot, allow us to clarify that the name comes from the fact that it *shuts out* the unnecessary light. A shutter lets in the amount of light required for your shot and also protects the image sensors because they could get damaged if overexposed.

A common misconception is that digital cameras don't have mechanical shutters at all. A digital camera could have a mechanical or electronic shutter – some cameras use both – but don't strain your eyes trying to spot them because some digital cameras (especially the ultra-portable types) don't have a shutter at all.

Mechanical Shutters:

A mechanical shutter uses a real (opaque) obstruction controlled by an electronic timer. There are different kinds of mechanical shutters:

a) Between-the-lens shutters: These are fitted inside the lens itself, usually near the aperture iris. Typically, this is made of several blades that slide over one another to open or shut the aperture. The greater the number of blades, the more circular is the aperture.

b) Focal-plane shutters: These shutters lie really close to